

CS202 Midterm – Real Exam Pattern Tracker & Practice

(Based on 9 student-shared paper reviews, Jun–Dec 2026 sessions)

This guide pulls out what is actually repeating across recent papers, so you practice the right things instead of everything equally.

1. What the reviews tell us about the paper

- Almost everyone calls the paper easy, and says it matches the handout + the "Web Dev Midterm Prep" video closely – not random outside material.
- MCQs (14): Mostly HTML & CSS, only 1–3 from JS. Direct concept/definition recall, not tricky.
- Short questions (2 × 3 marks): Very often "here is a code snippet, write its output," or "write a small CSS/HTML rule for a given scenario."
- Long questions (2 × 5 marks): Almost always a scenario + expected output is shown, and you write the HTML/CSS code that produces it. Tables, lists, hyperlinks, and image styling repeat a lot.

2. Exact MCQ facts confirmed by reviewers (memorize these)

- The first character encoding standard is ASCII.
- `toUpperCase()` converts a string to uppercase letters.
- `position: fixed` keeps an element in place even when the page is scrolled (it does not move).
- Colors can be specified in HTML/CSS 3 ways: Hexadecimal (`#RRGGBB`), RGB (`rgb(r,g,b)`), or Color name (e.g., `red`) – this is also the answer for "how is background color set."

3. Practice Set A – "Predict the Output" (Short Questions)

This is the single most repeated short-question type. Practice reading code and saying exactly what appears on screen.

Q1 (from a real paper):

```
<h2>Welcome,</h2>
```

```
<p>Hi, How are you?</p>
```

```
<a href="https://www.virtualuniversity.edu.pk">Virtual University Of Pakistan</a>
```

Output: A heading "Welcome," appears in large bold text, followed by a normal paragraph "Hi, How are you?", followed by a clickable underlined blue link reading "Virtual University Of Pakistan" – clicking it opens the VU website.

Q2 (from a real paper – the classic JS trap):

```
<p id="demo"></p>
```

```
<script>
```

```
var x = "3";
```

```
var y = 2;
```

```
var z = 1;
```

```
var sum1 = x + y;
```

```
var sum2 = y + z;
```

```
document.getElementById("demo").innerHTML = sum1 + "<br>" + sum2;
```

```
</script>
```

Output:

32

3

Why: x is a string ("3", in quotes), so x + y does concatenation, not addition → "3" + 2 = "32". But y and z are both plain numbers, so y + z = 2 + 1 = 3 (real addition). This exact "string + number vs number + number" trick has appeared more than once – practice it until it's automatic. *Hint: Quotes mein ho to text (jod diya jata hai), bina quotes ke ho to number (add hota hai).*

4. Practice Set B – CSS Pseudo-classes & Attribute Selectors

Q1 (exact paper question):

Write a CSS rule with a pseudo-class that changes the background color of the button to lightgreen on hover.

```
<button class="btn">Click the button</button>
```

Answer:

```
.btn:hover {
```

```
    background-color: lightgreen;
```

```
}
```

Q2 (exact paper question, paraphrased):

Three anchor tags are given, two contain "google" in their link, one doesn't. Write a CSS rule that makes only the anchor tags linking to "google" red and bold.

```
<a href="https://www.google.com">Search</a>
```

```
<a href="https://www.bing.com">Other</a>
```

```
<a href="https://mail.google.com">Mail</a>
```

Answer (using the [attribute*=value] "contains" selector):

```
a[href*="google"] {
```

```
    color: red;
```

```
    font-weight: bold;
```

```
}
```

*Hint: *= matlab "value ke andar kahin bhi yeh word ho" — yehi is qisam ke sawal ka jawab hota hai.*

5. Practice Set C — Build the HTML for a Given Output (Long Question)

This is the #1 most repeated long-question style: examiner shows you the rendered output (a picture), and you write the code.

Pattern seen repeatedly: two headings followed by an unordered list.

```
<h1>Main Heading</h1>
```

```
<h2>Sub Heading</h2>
```

```
<ul>
```

```
    <li>First item</li>
```

```
    <li>Second item</li>
```

```
    <li>Third item</li>
```

```
</ul>
```

Tip: Practice this exact skeleton (h1 + h2 + ul/li) by hand 4–5 times. Multiple reviewers say this almost word-for-word appeared as a long question.

6. Practice Set D — Tables

Two table styles have appeared:

(a) Simple numbered table (item 1, 2, 3...):

```
<table border="1">
```

```
<tr>
```

```
<th>Item No.</th>
```

```
<th>Name</th>
```

```
</tr>
```

```
<tr><td>1</td><td>Pen</td></tr>
```

```
<tr><td>2</td><td>Book</td></tr>
```

```
<tr><td>3</td><td>Bag</td></tr>
```

```
</table>
```

(b) URL parts table (break a URL into its components — Topic 26):

Given: <http://mydomain.com/myresource.html> — fill a table with the 5 parts.

Part	Value	Meaning
Scheme/Protocol identifier	http	type of internet service
Host	www (default)	domain host
Domain	mydomain.com	the internet domain name
Port	80 (default for http)	port number on host
Path/File name	myresource.html	the resource/

file on the server

Hint: scheme://host.domain:port/path/filename — yeh structure cram kar lo, table seedha isi se fill hota hai.

7. Practice Set E — Images: Hyperlinks & Styling

(a) Make an image act as a clickable link to a website (repeated request — "vu.jpg" → redirect to VU site):

```
<a href="https://www.vu.edu.pk">
```

```

```

```
</a>
```

Hint: ko ke andar daal do — pure image pe click karne se link khulta hai.

(b) Give an image a colored outline (one reviewer asked for a blue outline):

```
img {  
  
  outline-style: solid;  
  
  outline-color: blue;  
  
  outline-width: 3px;  
  
}
```

or shorthand:

```
img {  
  
  outline: 3px solid blue;  
  
}
```

(c) "Change a PNG's color to red" — bonus/uncommon type: This isn't directly in the official handout topics, so if it appears, the most likely intended answer (since transparent PNGs can be recolored with CSS filters) is:

```
img {
```

```
filter: brightness(0) saturate(100%) invert(13%) sepia(95%) saturate(6000%) hue-rotate(0deg);  
  
}
```

If the question seems simpler than this, it most likely actually means outline-color or background-color behind a transparent PNG — re-read the exact wording in the picture they show you, since "color of image" questions in this course almost always map back to outline/border/background, which you already know from Topics 38–39.

8. Priority Checklist Before Your Exam

1. Re-read Topics 14–17 (Links, Images, Tables, Lists) and Topics 38–39 (Box Model, Border/Outline) — these power most long questions.
2. Memorize the URL-parts table (Topic 02/26) — it has repeated almost every session.
3. Practice the string-concatenation JS output trap (Topic 56b) — it has repeated almost every session.
4. Drill :hover and [attribute*=value] selectors (Topics 44, 51) by writing them from memory 3–4 times.
5. Don't over-study advanced JS (loops, error handling, RegExp) — reviewers confirm JS contributes only 1–3 MCQs and rarely a long question.

Good luck — based on these reviews, if you're comfortable with HTML structure, basic CSS selectors, and the box model, you're already covering most of the paper.